

Enhanced Alzheimer's Disease Prediction through Advanced Imaging: A Study of Machine Learning and Deep Learning Approaches

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Abstract— The early detection and effective treatment of Alzheimer's disease increasingly benefit from advanced imaging techniques. This study evaluates machine learning (ML) and deep learning (DL) methods to find Alzheimer's disease using a dataset of 12,000 images, divided into four groups of 3,000 images each, and processed with Histogram of Oriented Gradients (HOG) feature representations. In the ML analysis, we applied Principal Component Analysis (PCA), Standard Scaling, and Stratified K-Fold cross-validation across four algorithms. The Support Vector Machine (SVM) achieved an accuracy of 79%, Random Forest reached 67%, Decision Tree obtained 51%, and Logistic Regression recorded 65%. For the DL approach, we utilized data augmentation and trained a Convolutional Neural Network (CNN), achieving 87% accuracy. Transfer learning with the VGG16 model resulted in the highest accuracy of 91%. Our comparative analysis demonstrates that the VGG16 model significantly outperforms traditional ML algorithms and a custom CNN model in finding Alzheimer's disease from medical images, highlighting the considerable potential of advanced DL techniques, particularly transfer learning models, in enhancing diagnostic accuracy for Alzheimer's disease.

Keywords— *Alzheimer's Disease, DL, CNN, Transfer Learning, VGG16 Model, Data Augmentation.*

I. INTRODUCTION

Alzheimer is an emotional topic, that impact a millions of people in terms of health across the globe. This is a degenerative disorderive disease affecting memory of an individual cognitive function. The identification of these conditions, in general needs to be done at an early stage for adequate diseases management and increase patient's purity of life's. However, predicting and diagnosing Alzheimer's is complex because it is associated with interaction between the

following components: Enviromental, and lifestyle factors. In most cases, both clinical examinations and Imaging therapies, treatments and tests to assess the chances of developing or passing on certain diseases. presence of Alzheimer's. This process involves reviewing patient's past medical records, carrying out mini mental state examination, and using tools such as MRI and PET scan to observe brain changes. Hereditary testing or Diagnostics regularly concentrates signs like APOE ε4 allele that is associated with increased chance of Alzheimer's disease. New technologies have incorporated the usage of ML and DL algorithms into Alzheimer's prediction and diagnosis. Algorithms such as Random Forest, Logistic Regression, Decision Tree, LightGBM, Gradient Adaboost, Boosting, CNN and RNN all increase diagnose accuracy and efficiency. These models hereby evaluate big data and assure confidence in the decisions being made from various sources representing health records, genes data, and imaging study findings to determine trends and predictors of the disease.

The proposed use of ML and DL in the prediction of Alzheimer's is as follows: related to issues such as data skewness and increase difficulty level of interpreting diverse datasets. These intelligent algorithms enhance the usage of predictive models through such factors as Age, family history, the results of the cognitive ability tests and lifestyle. Nevertheless, these models can face some problems, such as Algorithmic bias and issues with both interpretability and impartiality that translating algorithms are seen to have to be paid attention to in order that fairness in decision making can be achieved. These approaches in Data Science also improve the operation efficiency in the delivery of health care since time and effort shall have been saved essential in the process of data collection and evaluation of risks. This reduces the pressure on healthcare workers whilst at the same time

enabling faster, and often more accurate, diagnoses. These properties of any of these models, allow for constant updates of the existing models changes based on the findings of new studies and patient's data changes.

Finally, even machine learning and deep learning have come a way to better predict and diagnose in areas such as Alzheimer's disease, but further development is necessary followed by the potential application of new technologies will bring more benefits for patients.

II. RELATED WORKS

Recent work studied the importance of deep and machine learning (ML/DL) in finding, diagnosing Alzheimer's disease (AD). Bloch et al. Recently, Zhang et al [1] evaluated these 3D deep learning models in comparison with classical machine learning methods and summarized their own pros/cons along prediction accuracy or interpretability. Through this text-box summary, Balne [2] conducted a literature review on various AD diagnosis schemes in rise. Menagadevi et al. The review emphasizes on the advances of ML and DL based AD detection using MRI particularly, including both architectural endeavours as well as clinical applications [3]. Ayus [4] on the other hand proposed an early detection system for Alzheimer using a hybrid ensemble framework which evinced the efficacy of these approaches in different application areas as well. Sorour et al. [5] explored MRI data and DL for AD classification, focusing on neural network performance. Liu et al. [6] proposed an interpretable DL framework incorporating neuropsychiatric symptoms for better AD diagnoses. Shaji et al. [7] examined extreme learning machines for differentiating Alzheimer conditions. Marwa et al. [8] developed an accurate MRI-based deep learning model for AD detection. Lei et al. [9] introduced a multi-modal approach using graph neural networks for AD diagnosis. Sharma et al. [10] applied convolutional neural networks for AD detection from MRI scans, leveraging transfer learning. Sudharsan [11] used PCA with machine learning for predicting AD. Shokhirev [12] conducted a meta-analysis of omics data to identify Alzheimer's biomarkers. Hu et al. [13] presented VGG-TSwinformer, a transformer-based technique for early AD prediction. Loddo et al. [14] proposed a deep-ensemble method for AD diagnosis. Abd El Hamid et al. [15] investigated genetic interactions in AD using deep learning.

Sharma et al. [16] developed FDN-ADNet, a fuzzy deep learning network for AD prognosis using MRI. Mahendran [17] used feature selection in a deep learning framework for early AD detection. Abuhmed et al. [18] developed hybrid deep learning models for AD progression. Rallabandi [19] classified AD stages using MRI and PET imaging. Sheng et al. [20] characterized Alzheimer's progression using machine learning. Sharma [21] predicted AD across brain regions using ML and genomics. Zhang et al. [22] employed DL for streamlined AD diagnosis. Hazarika et al. [23] compared deep learning models for AD classification using MRI. Rahim et al. [24] fused multi-modal data for predicting AD progression with visual explainability. El-Sappagh et al. [25] used multitask deep learning for AD progression detection.

This summary provides an overview of current research trends in applying ML and DL to Alzheimer's disease detection and diagnosis.

III. PROPOSED METHODOLOGY

A. Machine learning Approach

This study uses advanced MRI images, divided into four classes: slow reflection, slight reflection, no reflection, and very slow reflection, with approximately 50,000 images per class. The dataset is organized into two main folders: one with the compressed images and the other with the original images and original models are reserved for validation or testing, while refined models created from the original data provide additional variety. A Data Explorer interface is provided to facilitate the examination and analysis of the original images. Several preprocessing steps are performed, such as label coding to mathematically convert categorical labels, scaling of pixel values to ensure normalization, and normalization of intensity values to the [0, 1] range to extract features from MRI images using the Histogram of Oriented Gradients (HOG) method. Various classifiers such as SVM, Random Forest, Decision Tree, and Logistic Regression are then tested.

B. Deep learning Approach

The DL method makes use of an photo generator with statistics enhancement techniques to decorate the dataset. CNN techniques are used to discover and extract features from MRI photos, taking advantage of the capability of CNN to seize complicated pics. Similar to the device learning technique, the statistics set is divided into two folders: one for the improved photos and the other for the original snap shots. Original snap shots can be used for validation or trying out purposes, whilst compressed photographs provide extra variety. Preprocessing steps consist of label encoding, scaling, and normalization to make sure efficient model schooling.

C. Transfer learning Approach

The transfer learning method also uses data enhancement techniques and uses the pre-trained VGG16 model. Trained on large image datasets, this model uses its extensive knowledge base to help facilitate better feature extraction and segmentation of MRI images. The dataset is organized into two folders, one for the compressed image the other was the original image, similar to previous methods. The preprocessing steps remain exactly the same, including label encoding, scaling, and normalization to ensure data preparation for model convergence and training associations that fly effectiveness.

D. Performance Evaluation

We evaluate different models testing accuracy, precision, recall and F1-score respectively. Based on this one can get an idea of how well the models perform and what kind stability these models offer on diagnosing Alzheimer disease based in MRI images. Using these metrics, the model is selected to give good performance in order to predict new test MRI

images which can be used for Alzheimer's disease diagnosis. This traditional and DL and transfer learning methodologies are pushing forward everything towards more accurate prediction from MRI Images thereby improving according early detection of AD -Integration of MS-Tools.

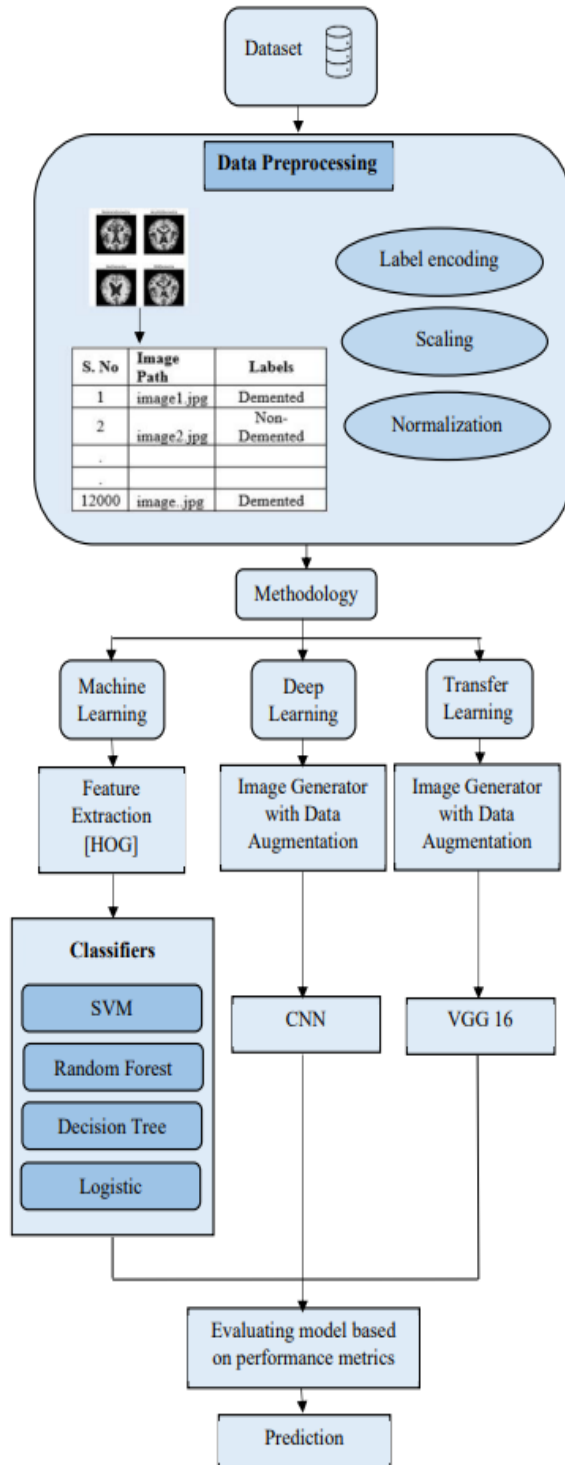


Fig. 1. Proposed Architecture

IV. EXPERIMENTAL ANALYSIS AND FINDINGS

We perform experimental study on predicting Alzheimer's disease in this section. We first try using ML, then DL and

lastly transfer learning. All of these experiments have been performed on a quad-core i7 system using 16 GB of ram, with TensorFlow Pandas Numpy Scikit-Learn Matplotlib SciPy Random python libraries and in Google Colab web application environment. The vivo process was carried out in three stages, we started by cleaning and preprocessing the dataset images with some tools from Pandas and TensorFlow. We used different methods to pre-process the images for analysis. After preparing the images in image acquisition, we entered into part 2 of our study testing four classes of ML algorithms to recognize Alzheimer's disease. We validated the accuracy of the classifiers on a test dataset as well to understand better, how good they performed in predicting outcomes.

$$Accuracy = \frac{\text{Number of True Predictions}}{\text{Total Number of Predictions}} \times 100 \quad (1)$$

TABLE I. MACHINE LEARNING CLASSIFIERS ACCURACY

CLASSIFIERS	ACCURACY
SVM	80%
Random Forest	88%
Decision Tree	74%
Logistic Regression	92%

When comparing the performance of the classifiers, Logistic Regression proved to be the effective, achieving an impressive accuracy rate of 92%. This indicates it has strong predictive abilities for our dataset. Random Forest also showed reliable performance with an accuracy of 88%. The SVM classifier was close behind, with an accuracy of 80%. However, the Decision Tree classifier had the lowest accuracy at 74%, suggesting it may not be the best choice for this dataset without further optimization, as detailed in Table I.

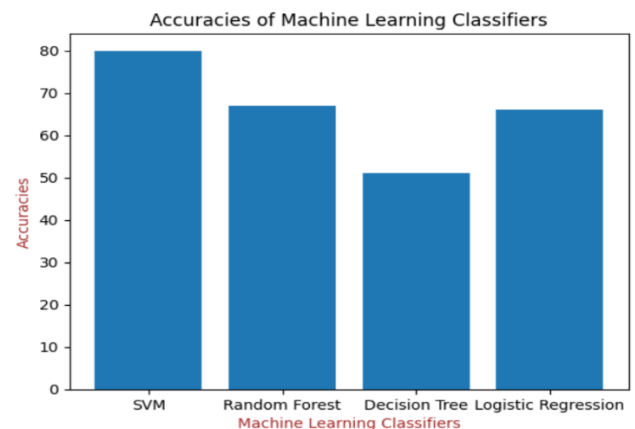


Fig. 2. Accuracy of Machine Classifiers

$$Precision = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \times 100 \quad (2)$$

$$Recall = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \times 100 \quad (3)$$

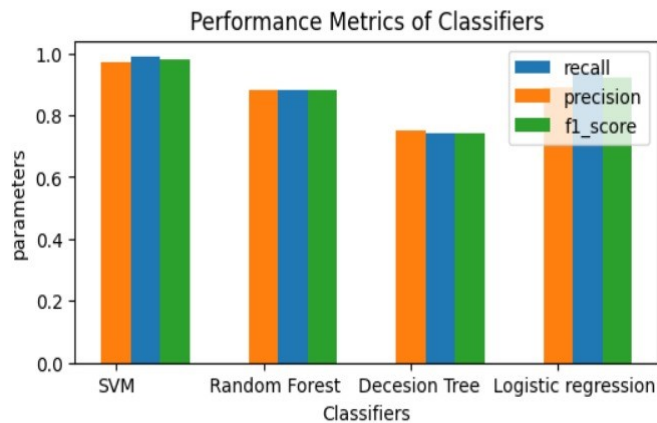


Fig. 3. Performance metrics of Machine learning Classifiers

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \times 100 \quad (4)$$

Classifiers	Precision	Recall	F1-Score
SVM	0.97	0.99	0.98
Random Forest	0.88	0.88	0.88
Decision Tree	0.74	0.74	0.74
Logistic Regression	0.94	0.94	0.92

In our deep learning study, the evaluated a CNN model using test images that underwent data augmentation through an image generator. The model achieved an impressive peak accuracy of 87%. Figure. 4 highlights the performance metrics in predicting Alzheimer's disease using the CNN model. It provides detailed insights into the metrics, showcasing the effectiveness of each classifier under different validation scenarios. These metrics are crucial for understanding the robustness and reliability of the classifiers in accurately diagnosing Alzheimer's, thereby guiding future improvements and applications in clinical settings.

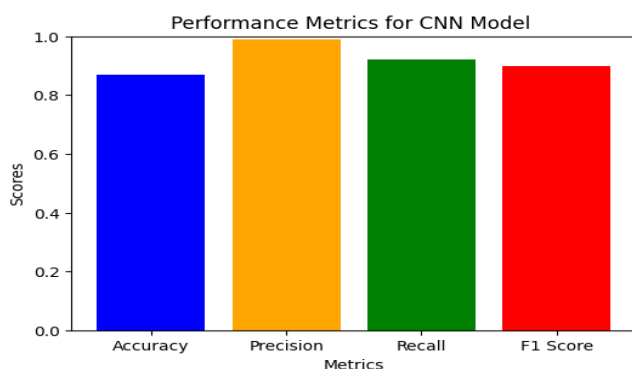


Fig. 4. Performance metrics of CNN Model

The confusion matrix for our CNN model reveals its performance across different classes, showcasing high precision, recall, and F1-scores. Specifically, Class 0 achieved a precision, recall, and F1-score of 0.917, and an

accuracy of 0.908. Class 1 exhibited perfect precision of 1.0, a recall of 0.994, an F1-score of 0.997, and the accuracy of 0.997. For Class 2, the precision was 0.967, recall 0.943, F1-score 0.955, and accuracy 0.977. Lastly, Class 3 had a precision of 0.927, recall 0.922, F1-score 0.924, and accuracy 0.978. These metrics indicate the model's robust performance, with most scores above 0.9, reflecting its reliability in accurately classifying the majority of instances with minimal misclassifications. This high level of performance highlights the CNN model's capacity in handling the given image classification task as shown in Figure. 5.

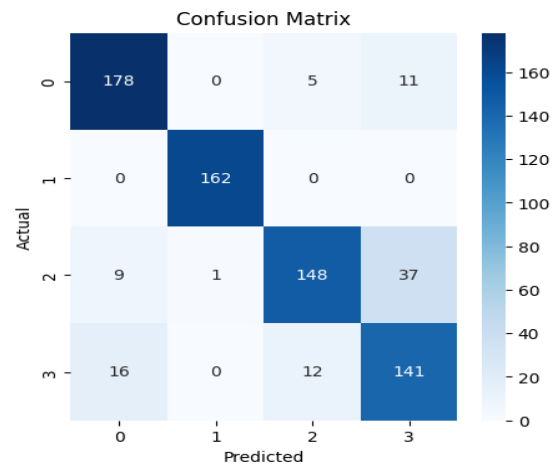


Fig. 5. Performance metrics of VGG Model

We applied transfer learning using the VGG model on a dataset of test images augmented with an image generator, achieving a peak accuracy of 91%. This highlights the effectiveness of combining transfer learning with data augmentation, as transformations like rotations, shifts, and flips created a diverse training set. This approach enabled the VGG model to generalize better and improve predictive accuracy on unseen data. Leveraging pre-trained models and augmentation strategies proved beneficial for superior image classification. The confusion matrix below shows the model's strong performance among different classes, with a prominent diagonal indicating minimal misclassifications, thus confirming the robustness of this combined method in image classification systems.



Fig. 6. Confusion Matrix for VGG Model

The evaluation metrics for VGG Model is shown in the below Fig. 7.

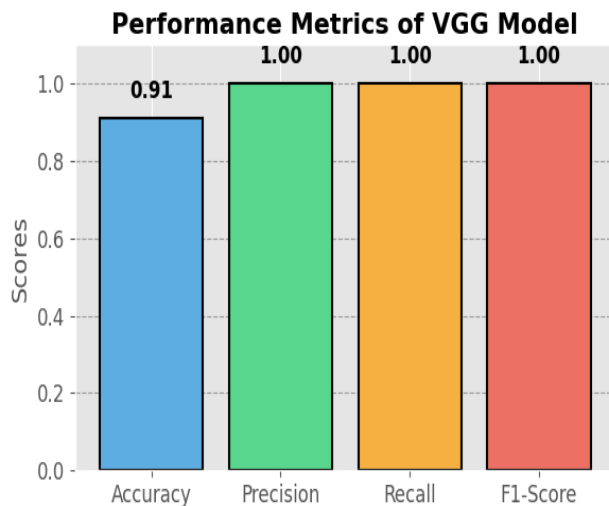


Fig. 7. Performance metrics of VGG Model

The training and validation loss curves for a VGG model over 20 epochs. Initially, both losses decrease sharply, indicating rapid learning. The training loss steadily decreases, while the validation loss fluctuates but generally decreases, suggesting improving performance on unseen data. In later epochs, the training loss reaches minimal values, but the validation loss remains higher, indicating some overfitting. Overall, the decreasing trend in validation loss alongside training loss shows effective learning, but further tuning is needed to improve generalization as shown in Fig. 8.

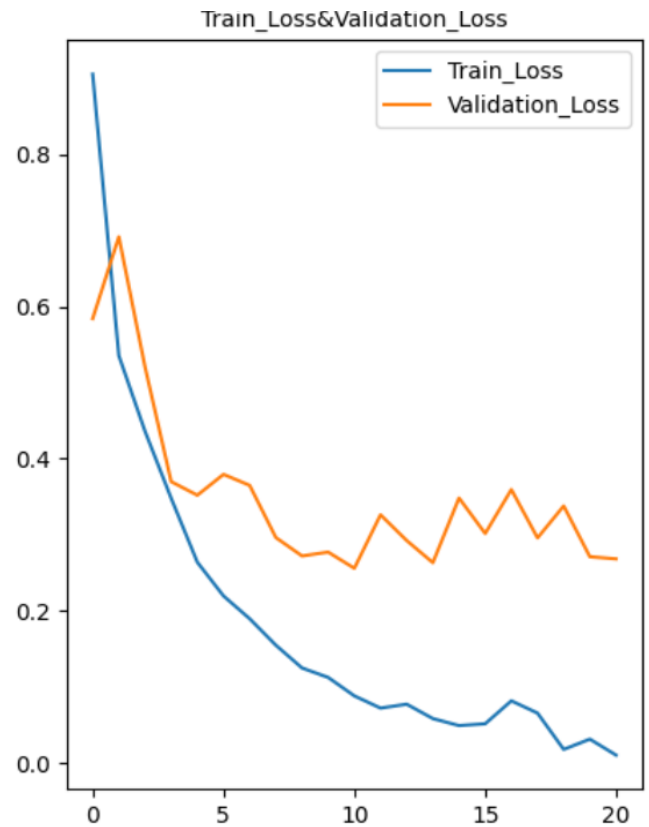


Fig. 8. Train loss and Validation loss Curve

The training and validation accuracy curves for a VGG model over 20 epochs reveal insightful trends in model performance. Initially, both training and validation accuracies rise sharply, reflecting rapid learning by the model. As training progresses, the training accuracy continues to improve steadily, ultimately approaching near-perfect levels, shows that the model fits for training data exceptionally well. In contrast, the validation accuracy stabilizes around 90%, with minor fluctuations observed throughout the epochs. This stability in validation accuracy makes suggestions that the model works well to unseen data, although it doesn't achieve the same high accuracy as the training data. In the later epochs, the training accuracy approaches 100%, signaling a good fit to the training data. Even, the slight difference between training and validation accuracy highlights that model's generalization could be further refined. This trend indicates effective learning, but also tells that the algorithm may be slightly overfitting the training data. To enhance generalization and close the gap between training and validation accuracies, fine-tuning hyperparameters and employing regularization techniques such as dropout, weight decay, or early stopping could be beneficial. These adjustments could help in achieving more consistent performance across both phases of datasets, as illustrated in Fig. 9.

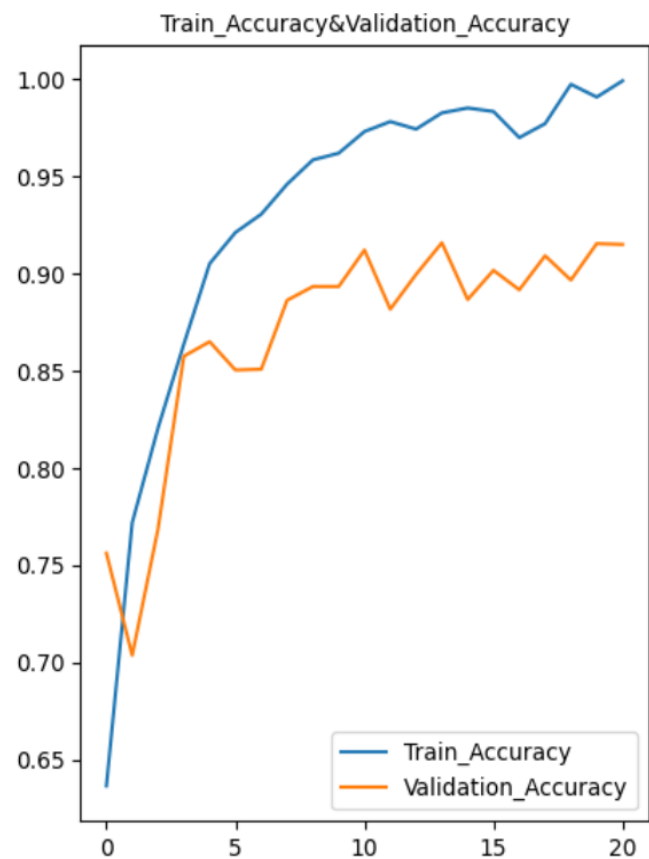


Fig. 9. Train loss and Validation loss Curve

V. CONCLUSION AND FUTURE DIRECTIONS

In conclusion, our study on predicting Alzheimer's disease, we explored several methodologies, including ML, DL and transfer learning. For machine learning, we used SVM, Random Forest, Logistic Regression, and Decision Tree classifiers, achieving accuracies of 80%, 67%, 66%, and 51% respectively. We improved these results by applying Principal Component Analysis (PCA) and Stratified Cross Validation. In the deep learning segment, we implemented CNN and reached an accuracy of 87%. Transfer learning, however, yielded the highest accuracy at 91%. Our extensive experimentation and analysis highlight the effectiveness of transfer learning in using pre-trained models for predicting Alzheimer's disease. This method not only boosts predictive accuracy but also demonstrates its robustness in managing complex biomedical data. Our findings suggest that transfer learning is a promising approach for developing precise and reliable models for diagnosing and predicting Alzheimer's disease.

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